

# Hoa Huynh

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**LinkedIn:** [linkedin.com/in/huynhthaihoa](https://linkedin.com/in/huynhthaihoa)

**Portfolio:** [huynhthaihoa.github.io](https://huynhthaihoa.github.io)

**GitHub:** [github.com/huynhthaihoa](https://github.com/huynhthaihoa)

**LeetCode:** [leetcode.com/u/huynhthaihoa](https://leetcode.com/u/huynhthaihoa)

## ABOUT

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Computer Vision Engineer with 7+ years in the field, including 5+ in industry developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. Experienced in 2D/3D perception, inference optimization, and multi-sensor integration. Interested in Visual SLAM, 3D vision, and robotics.

## SKILLS

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**Programming Languages:** C, C++, Python, C#, MATLAB

**Perception & Deep Learning:** Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet

**Edge AI Deployment:** TI Edge AI/TIOVX (SK-TDA4VM, TDA4AEN), NVIDIA Jetson Nano/Xavier NX (TensorRT, DeepStream), Samsung Exynos V920, ONNX Runtime, TFLite

**Sensors & Calibration:** RGB/IR pinhole/fisheye, FLIR thermal, Intel RealSense, LUCID Helios2 ToF, Ouster LiDAR, ROS

**MLOps & Tooling:** Git, SVN, Docker, CMake, TensorBoard, DVC, W&B, MLflow, Jira, Confluence

**Databases:** Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

**Additional Languages & Frameworks:** Node.js, React, JavaScript, Java, LaTeX, WPF, Qt, FFmpeg

**Spoken Languages:** English (professional), Korean (elementary), Vietnamese (native)

## EXPERIENCE

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### DeltaX

Seoul, South Korea

*Development Team Lead & AI Researcher*

*June 2022 – July 2026*

- Led an R&D team from December 2023, driving the architecture, technical roadmap, and end-to-end delivery of an In-Cabin Monitoring System (ICMS) that aligned with Euro NCAP standards, while coordinating module development, milestone tracking, and client demonstrations across cross-functional teams.
- Architected a real-time edge AI pipeline using a single RGB/IR fisheye camera, C/C++ inference on an SoC (System-on-Chip), and a Node.js/React web visualization UI.
- Integrated 8 deep learning models (object detection, 2D/3D body keypoint estimation, facial landmark detection, gaze estimation, and image classification), achieving an average of 14 FPS on an 8-TOPS edge platform and 9 FPS on a 4-TOPS edge platform.
- Delivered 4 Proof-of-Concept (PoC) projects for 2 Korean automotive vendors within 3–9-month timelines: 42dot (Phase 1: January – June 2024; Phase 2: August – October 2024), Pyeong Hwa Automotive (Phase 1: May 2024 – January 2025; Phase 2: July – October 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC targets, deploying on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Developed the 3D perception stack for worker localization, 3D display, and ADAS applications (Advanced Driver-Assistance Systems): monocular depth estimation (with pseudo-LiDAR conversion) trained under LiDAR/RGB-D supervision, semantic segmentation, visual SLAM, and real-time multi-camera stitching from fisheye and pinhole inputs.
- Designed a slicing-based preprocessing scheme derived from SAHI (“Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection”) to raise recall on small objects for YOLOv8-p2/YOLOv9n models while maintaining real-time inference latency.
- Built the team’s multi-sensor calibration, data collection, and analysis tooling (ROS, MATLAB, Python), and assembled real-world ADAS/in-cabin datasets across diverse subjects and lighting conditions.

## Smartdeep

Research Engineer

Seoul, South Korea

May 2021 – May 2022

- Built a 16-channel CCTV surveillance system in C++ (OpenCV, FFmpeg) with a C# WPF front end, aligned with KISA Intelligent CCTV certification requirements for moving- and abnormal-object detection, recording, and event search.
- Owned dataset curation, training, and validation pipelines for Darknet YOLOv4 detection models.

## Ubay Solution

AI Research Intern

Seoul, South Korea

December 2020 – April 2021

- Developed an OCR system for Busan Port to read truck license plates and container codes, benchmarking SSD, YOLOv4, and Faster R-CNN on a custom dataset expanded via augmentation and web crawling.
- Built a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano/Xavier NX for real-time inference.

## ANDA Lab – Soongsil University

Research Assistant

Seoul, South Korea

September 2018 – August 2020

- Diagnosed failure modes in a taillight-based vehicle localization system via experimental analysis in Python, OpenCV, and MATLAB, then designed the algorithms to address them: overlapping-LED detection for vehicular visible light communication (VLC), LED taillight detection for optical camera communication (OCC), and stereo-vision taillight matching and pairing for vehicle-to-vehicle positioning.
- First-authored 3 peer-reviewed papers on camera-based nighttime vehicle positioning, including IEEE Access (SJRN Q1) and MDPI Applied Sciences (SJRN Q2).

## Global Cybersoft JSC

Software Engineer Trainee

Ho Chi Minh City, Vietnam

March 2017 – August 2017

- Built C++ plugins for Japanese clients: MFC plugins for SCREEN Holdings' industrial printer system, and Adobe Illustrator SDK plugins (C++, JavaScript) for an advertising agency.

## EDUCATION

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### Soongsil University

Master of Engineering, Information and Communication Technology

Seoul, South Korea

September 2018 – August 2020

- **GPA:** 4.01/4.5
- **Thesis:** [LED Taillight Matching Algorithm for Nighttime Vehicle Positioning](#) (written in English)
- **Supervisor:** Professor Myungsik Yoo

### University of Information Technology – Vietnam National University

Bachelor of Engineering, Software Engineering

Ho Chi Minh City, Vietnam

August 2013 – June 2018

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** [Blood Vessel Segmentation on DICOM Images](#) (written in Vietnamese)
- **Supervisor:** Thi-Vuong Pham, M.Sc.

## PUBLICATIONS

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(*J*: International Journal, *W*: International Workshop, *K*: Korean Journal/Conference)

**K2.** Uddin S M Nadim et al. (including **Huynh Thai Hoa**), “A Railway Framework for Person (Worker) Localization based on Depth Estimation and GPS”, *The Korean Institute of Defense Technology Fall Conference* (2024). [Reference](#)

**W1.** Jamie Spencer et al. (including **Huynh Thai Hoa**), “The Second Monocular Depth Estimation Challenge”, *The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (2023). [Reference](#)

**J2.** **Thai-Hoa Huynh**, Myungsik Yoo, “A Taillight Matching and Pairing Algorithm for Stereo-Vision-Based Nighttime Vehicle-to-Vehicle Positioning”, *MDPI Applied Sciences* (2020). [Reference](#)

**K1.** **Thai-Hoa Huynh**, Myungsik Yoo, “Nighttime LED Taillight Detection Method with Optical Camera Communication System”, *The Journal of Korean Institute of Communications and Information Sciences* (2020). [Reference](#)

**J1. Thai-Hoa Huynh**, Tuan-Anh Pham, Myungsik Yoo, “Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System”, *IEEE Access* (2019). [Reference](#)

## SELECTED PROJECTS

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**State Estimation Simulations:** Python simulations comparing how a robot estimates its pose from noisy motion and sensor data: recursive filtering (EKF, Invariant EKF, UKF) vs. batch and incremental optimization (factor graphs, iSAM, bundle adjustment), built from scratch with Lie-group math. [GitHub](#)

**Monocular Metric Depth Estimation:** PyTorch pipeline estimating per-pixel metric depth from a single image, with pluggable encoders and decoders, trained on NYUv2/KITTI datasets with EdgeAI-TIDL-based quantization-aware training support and deployed via ONNX/TFLite export. [GitHub](#)

**Notch-Based Point Cloud Registration:** C++ point-cloud registration aligning 3D scans to a reference model via notch-keypoint correspondence matching, for industrial inspection and quality verification. [GitHub](#)

**LiDAR-Based Ego Lane Detection:** classical signal-processing pipeline (ground extraction, intensity analysis, polynomial fitting) for interpretable ego-lane detection directly from LiDAR point clouds. [GitHub](#)

*(Full project list available on portfolio: [huynhthaihoa.github.io/#projects](https://huynhthaihoa.github.io/#projects))*

## SELECTED CERTIFICATIONS

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**Coursera:** Introduction to Machine Learning in Production.

**Hugging Face:** Fine-tuning Language Models, AI Agents Fundamentals, Fundamentals of Group Relative Policy Optimization.

*(Full certification list available on portfolio: [huynhthaihoa.github.io/#certifications](https://huynhthaihoa.github.io/#certifications))*